

6.2 User Request Modularity

Next, we varied the *user request modularity* (number of attributes per atomic service) according to the available distributions, expecting the *sophistication* of the agents to be distributed according to the modularity of the user requests, i.e. the more complicated (in terms of modular non-reducible tasks) the user request, the more sophisticated (in terms of the number of attributes) the agents needed to fulfil it.

We first applied the Uniform distribution as a control, and graphed the results in Figure 9. The *observed* frequencies for the number of agent attributes mostly matched the *expected* frequencies, which was confirmed by a χ^2 test; with a *null hypothesis* of *no significant difference* and *ten degrees of freedom*, the χ^2 value was 1.049, below the critical 0.95 χ^2 value of 3.940.

We then applied the Gaussian distribution as a reasonable assumption for the behaviour of a large group of users, and graphed the results in Figure 10. The *observed* frequencies for the number of agent attributes again followed the *expected* frequencies, but there was variation which led to a failed χ^2 test; with a *null hypothesis* of *no significant difference* and *ten degrees of freedom*, the χ^2 value was 50.623, not below the critical 0.95 χ^2 value of 3.940.

Finally, we applied the Power Law distribution to represent a relatively extreme variation in user behaviour, and graphed the results in Figure 11. The *observed* frequencies for the number of agent attributes also followed the *expected* frequencies, but there was variation which led to a failed χ^2 test; with a *null hypothesis* of *no significant difference* and *ten degrees of freedom*, the χ^2 value was 61.876, not below the critical 0.95 χ^2 value of 3.940.

In all the experiments the *observed* frequencies of the number of agent attributes followed the *expected* frequencies, with some variation in two of the experiments. This could only be confirmed statistically, by a χ^2 test, for the Uniform distribution experiment. In the Gaussian and Power distribution experiments the χ^2 tests failed by considerable margins, most likely because the evolving agent populations were still self-organising to match the user behaviour, shown by the *observed* frequencies *approaching* the *expected* frequencies, but not yet sufficiently to meet χ^2 tests, because by the thousandth *time step* (user request event) each user had placed an average of only ten requests.

Collectively, the experimental results confirm that the self-organised *diversity* of the evolving agent populations is relative to the *selection pressures* of the user base, which was confirmed statistically for most of the experiments. So, we have determined an effective understanding and quantification for the self-organised *diversity* of the evolving agent populations of our Digital Ecosystem. While the minor experimental failures, in which the Digital Ecosystem responded more slowly than in the other experiments, have shown that there is potential to optimise the Digital Ecosystem, because the evolutionary self-organisation of an ecosystem is a slow process [3], even the accelerated form present in our Digital Ecosystem.

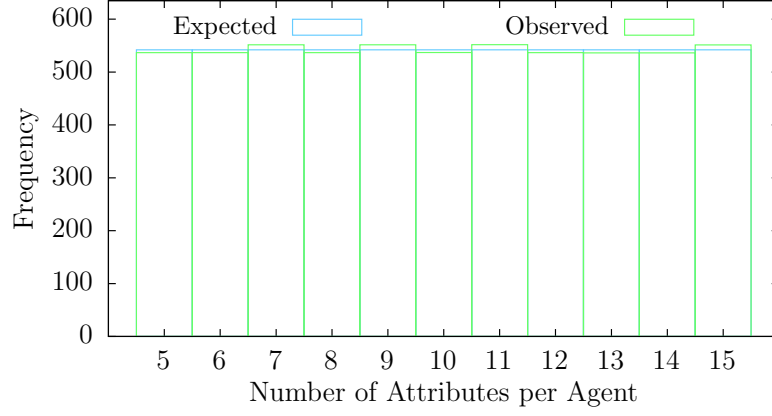


Figure 9: *Graph of Uniformly Distributed Agent Attribute Frequencies: The observed frequencies for the number of agent attributes mostly matched the expected frequencies.*

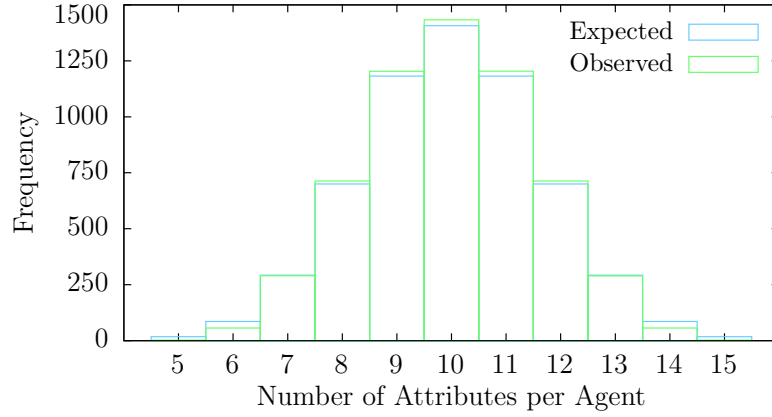


Figure 10: *Graph of Gaussian Distributed Agent Attribute Frequencies: The observed frequencies for the number of agent attributes again followed the expected frequencies, but there was variation.*

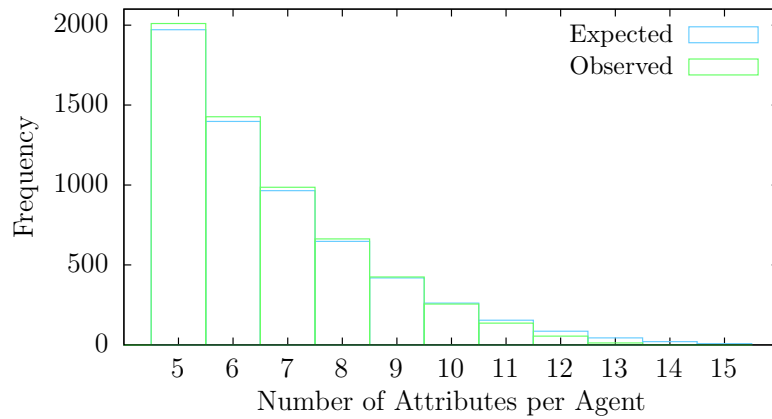


Figure 11: *Graph of Power Law Distributed Agent Attribute Frequencies: The observed frequencies for the number of agent attributes also followed the expected frequencies, but there was variation.*

7 Conclusions

We have confirmed the fundamentals for a new class of system, Digital Ecosystems, created through combining understanding from theoretical ecology, evolutionary theory, MASs, distributed evolutionary computing, and SOAs. Digital Ecosystems, where the word *ecosystem* is more than just a metaphor, being the digital counterpart of biological ecosystems, and therefore having their desirable properties, such as scalability and self-organisation. It is a complex system that shows emergent behaviour, being more than the sum of its constituent parts.

With the core infrastructure of Digital Ecosystems defined, some intelligence-based augmentations have been considered for its acceleration and optimisation [7, 4]. In which the agents were endowed with memory, intelligence and the ability to share information, with the aim of optimising the ecological dynamics, through the emergent optimisation of the agent migration patterns. While not yet as sophisticated as swarm intelligence, we would argue it is a step in that direction, and we would like to investigate further in the future.

The ever-increasing challenge of software complexity in creating progressively more sophisticated and distributed applications, makes the design and maintenance of efficient and flexible systems a growing challenge [30], for which current software development techniques have hit a *complexity wall* [29]. In response we have created Digital Ecosystems, the digital counterparts of biological ecosystems, possessing their properties of self-organisation, scalability and sustainability [25]; Ecosystem-Oriented Architectures that overcome the challenge by automating the search for new algorithms in a scalable architecture, through the evolution of software services in a distributed network.

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